Utkarsh Sarvjeet Tiwari

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Experience

Centre for Development of Advanced Computing (C-DAC) — **Pune, India**
HPC-Quantum Computing Research Fellow — *July 2026 - Present*

- Leading a research team on DFT-in-active-space embedding with VQE, integrating classical DFT with quantum solvers for scalable hybrid HPC-quantum molecular simulation.
- Migrated an IBM/Qiskit-based VQE pipeline to Quantinuum hardware via pytket, resolving cross-framework operator-mapping and qubit-ordering incompatibilities.
- Manage execution across real QPUs - IBM, Quantinuum, and AWS Braket (Rigetti) - with HPC cluster access for coupled classical/quantum workloads.
- Diagnosed a critical pass-manager ordering bug causing incorrect noise-model transpilation in the benchmarking pipeline.

NLIP Lab, Indian Institute of Technology Hyderabad — **Sangareddy, India**
Research Intern (NLP) — *Jan 2026 - June 2026*

- Conducted AI and data evaluation of culturally adapted mathematical content, assessing quality, cultural relevance, language, social context, realism, and coherence.
- Built an information retrieval pipeline for academic papers using RAG, ChromaDB, Elasticsearch, and Sentence Transformers, with structured extraction of key findings, limitations, citations, tables, and images.
- Developed data pipelines using Docling for document parsing and PostgreSQL for structured storage; implemented citation graph embeddings.
- Performed retrieval evaluation and optimization across hybrid dense and sparse search strategies.

Aryahs World Infotech (OPC) Pvt. Ltd. — **Navi Mumbai, Maharashtra**
AI/ML Intern — *Dec 2025 - Jan 2026*

- Developed a WhatsApp automation system for the marketing team, including job scheduling and privacy control features.
- Built an ML-based symptom classifier mapping user inputs to probable medical conditions.
- Contributed full-stack features in Python, Flask, MySQL & JavaScript, with real-time datasets, testing, and Agile sprints.

CybraneX Pvt. Ltd. (Industry-Defined Research, under Faculty Mentorship) — **Remote**
Researcher — *Oct 2025 - June 2026*

- Led a 3-member research team and was a major contributor to a hybrid quantum-classical machine learning pipeline for mineral anomaly detection (lithium and vanadium deposits) using NASA EO-1 Hyperion hyperspectral satellite imagery, benchmarking multiple detection methods including a quantum autoencoder-based approach.
- Part of the team behind **TerraQ**, built to showcase our solution and problem statement: lithium/vanadium anomaly maps from a 6-qubit quantum autoencoder pipeline. Worked on the backend and dataset. Live

Education

Shah and Anchor Kutchhi Engineering College — **Mumbai, Maharashtra**
B.Tech in Electronics and Computer Science Engineering - CGPA: 8.17/10.0 — *Sept 2023 - 2027 (Expected)*

Skills

AI & Retrieval: NLP, Retrieval-Augmented Generation (RAG), Information Retrieval, Hybrid Search, BM25, Reranking, LLM Evaluation
Languages: Python, C, C++, JavaScript, TypeScript
Frameworks & Libraries: FastAPI, Flask, React, Sentence Transformers, Scikit-learn, TensorFlow
Data & Infrastructure: PostgreSQL, MySQL, MongoDB, Qdrant, ChromaDB, Elasticsearch, Docker, Git
Quantum Computing: Qiskit, pytket, PennyLane, VQE; IBM and Quantinuum QPUs, Rigetti via AWS Braket

Projects

Litmus - RAG Strategy Evaluation Workbench | *Python, FastAPI, React, TypeScript, Qdrant, Docker* | GitHub

- Built Litmus, a RAG evaluation workbench comparing 3 strategies-hybrid, graph-based, and agentic-over the same document corpus through a unified API.
- Engineered hybrid retrieval using dense search, BM25, reciprocal rank fusion (RRF), and cross-encoder reranking; added server-side citation validation and refusal handling.
- Developed an evaluation pipeline with Recall@K, nDCG, and a versioned LLM judge; implemented configuration-aware regression checks and tracked failed judge calls separately from scored results.

vaak - Local Multilingual Voice AI Agent | *Python, FastRTC, FastAPI, Ollama, Faster-Whisper, Kokoro, Docker* | GitHub

- Built a fully local, full-duplex WebRTC voice agent (no cloud APIs): sentence-chunked LLM tokens stream into Kokoro TTS; English routed to Moonshine, other languages to int8 Faster-Whisper.
- Per-session isolation, semaphore-capped concurrency, barge-in cancellation, Ollama retry with spoken fallback; 104 tests (73%).

Awards & Hackathon Experience

- **Finalist:** Innovate-a-thon (IIT ISM), HackWithUttarPradesh '25; **Grand Finale:** HackSetu 1.0; **Paper Selection:** E^2 Calonics'25.